GLOBAL AI CONSTRUCT · BRISBANE · MON 31 AUG 2026

Small Models, Sharp Jobs.

Build a local agent that proves its work.

85 minutes · build along or observe **Offline** · your runtime, your machine

Victorian crash sample · attributed, de-identified

JK

BUILDER / SPEAKER / COMMUNITY CONTRIBUTOR

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Agentic AI, Azure AI, coding-agent workflows, .NET and local models that have to survive a bad Wi-Fi night.

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A workflow you can see.

0–8	**Welcome and setup**	Explain the recovery lane; attendees build or observe.
8–16	**Hello and typed JSON**	Prove a local runtime can answer, then show a no-tool typed contract.
16–30	**Gather**	Filter the approved crash sample by date, term and cap. Show a supported and no-evidence result.
30–45	**Extract**	Send the question and compact pack only. Inspect selected IDs, rationale and confidence.
45–57	**Analyse**	Send only code-validated selected records.
57–68	**Gates and workflow**	Show invalid IDs and low confidence stopping safely; run the identical linear workflow.
68–77	**Recovery and checkpoints**	Pairing, saved Gather output, or a checkpoint solution.
77–85	**Close and hidden extras**	Model comparison, MCP and Harness only if time remains.

BEFORE ANYTHING RUNS

Build along, or observe. Both work.

```
dotnet test
dotnet run --project src/Workshop.App -- gather --term intersection
```

BUILDERS

Start in `starter/`

Your own OpenAI-compatible endpoint: MAF_ENDPOINT, MAF_MODEL, MAF_API_KEY.

OBSERVERS

No setup required

Every checkpoint has a saved run on screen. You still see the whole trust boundary.

STUCK

Five minutes, then move

Pair up, copy the stage from `solution/`, or watch. Do not debug in public.

THE WHOLE WORKSHOP ON ONE SLIDE

Smaller context. Sharper work.

Gather → Extract → gate → Analyse → gate

Deterministic Gather narrows the corpus. Extract turns a question plus a capped pack into typed record IDs. Code validates before Analyse ever runs. **Teal steps are ordinary C#** — no model gets a vote there.

UP NEXT

Prove the lane, then give it a **contract.**

A successful hello is a checkpoint, not a product.

`minutes 8–16`

02

CP-01 · LOCAL HELLO

One prompt. One answer. No tools.

```
MAF_ENDPOINT=http://localhost:1234/v1
MAF_MODEL=<the model you already have loaded>
MAF_API_KEY=<only if your runtime demands one>

dotnet run --project src/Workshop.App -- smoke
```

ACCEPTANCE `smoke` echoes the exact token; a down endpoint or unloaded model exits non-zero.

CP-02 · TYPED JSON

Useful output needs a shape.

```
dotnet run --project src/Workshop.App -- typed
```

MODEL

Returns a small typed contract

A handful of fields. Not prose, not a tool call.

CODE

Parses and validates it

Malformed output is rejected here, never quietly accepted.

BOUNDARY

No tools in this call

The local path deliberately does not combine tools and typed JSON.

ACCEPTANCE `typed` prints raw JSON then the parsed contract; malformed output exits non-zero.

UP NEXT

Evidence first.
Reason later.

The first real job is not a model call at all.

`minutes 16–30`

03

CP-03 · GATHER

A bounded query, not arbitrary files.

```
dotnet run --project src/Workshop.App -- gather --term intersection
dotnet run --project src/Workshop.App -- gather --term cyclist
# the second command is the no-evidence branch
```

ACCEPTANCE Code filters the approved crash sample by date, term and cap — then a query that returns nothing.

WHY THIS STAYS DETERMINISTIC

The model may ask. It may not reach.

MAY

Date, term, cap

Three parameters over one approved Victorian crash sample.

MAY NOT

Name a path

No filesystem, no folder listing, no unbounded result set.

RESULT

A compact pack

Capped, attributed records — the pack goes forward, the dataset does not.

RECOVERY `workshop/reference-run/gather-intersection.json` is the expected pack if a machine is not ready.

UP NEXT

Ask the model to **select**, not to conclude.

Question plus capped pack in. Typed record IDs out.

`minutes 30–45`

04

CP-04 · EXTRACT

Selected IDs, rationale, confidence.

```
dotnet run --project src/Workshop.App -- run --term intersection

extract JSON: { "selectedIds": [...], "confidence": 0.0–1.0 }
```

ACCEPTANCE Unknown IDs, duplicate IDs and malformed output are rejected in code — not argued with in the prompt.

UP NEXT

High effort on a small set.

Only records that already passed the gate get analysed.

`minutes 45–57`

05

Only validated records reach this call.

IN

Code-validated selection

The compact set Extract chose and validation accepted. Nothing else.

EFFORT

Highest of the three

Low effort finds candidates; medium structures them; high explains the small set.

OUT

A grounded finding

Formatted deterministically for an analyst — never a claim decision.

ACCEPTANCE Low confidence takes the caution branch instead of producing a confident answer.

UP NEXT

The model proposes.
Code decides.

Then the identical sequence, as a reusable workflow.

`minutes 57–68`

06

THREE VISIBLE OUTCOMES

Every branch is a success.

NO EVIDENCE

Stop early

Gather returned nothing. Extract and Analyse never run.

INVALID OR LOW CONFIDENCE

Caution result

Unknown ID, malformed JSON, or confidence under the threshold.

SUPPORTED

Grounded analysis

Validated records, an explained finding, and the evidence still on screen.

ACCEPTANCE Invalid IDs and low confidence stop safely — a refusal is a correct answer.

CP-06 · WORKFLOW

Same sequence. Visible topology.

```
dotnet run --project src/Workshop.App -- run --term intersection
dotnet run --project src/Workshop.App -- workflow --term intersection
# explicit calls, then the same sequence as one reusable workflow
```

ACCEPTANCE No evidence bypasses Extract and Analyse. The workflow adds reuse and topology, not intelligence.

UP NEXT

Catch up without catching fire.

Pairing, saved output, or a checkpoint solution.

minutes 68–77

07

PICK A LANE AND REJOIN

Three ways back into the room.

PAIR

Sit with a working machine

One laptop per pair is enough to hit every checkpoint.

SAVED OUTPUT

Use the reference run

`workshop/reference-run/gather-intersection.json` is the real bounded pack.

SOLUTION TREE

Copy one stage

`solution/` is the explicit catch-up state. Take the stage, rejoin at the next handoff.

NOTE A hosted OpenAI-compatible endpoint is participant-owned and optional. Never share a facilitator key.

UP NEXT

Close and hidden extras.

Model comparison, MCP and Harness only if time remains.

`minutes 77–85`

08

THE PART THAT OUTLIVES THE MODEL

Teach the boundary, not the benchmark.

Models change every few months. Constrained inputs, reduced context, validated output and a real recovery path do not. **The model proposes; code decides** — that sentence still holds when tonight's model is three generations old.

THANK YOU

Small jobs. Clear evidence. Provable outcomes.

What would make this easier to follow, or safer to reuse on Monday?

SLIDES AND CHECKPOINTS

`small-models-sharp-jobs-workshop-20260829.surge.sh/slides/`

Deck, agenda, commands and the saved reference run.